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Paper Code : 21327

F–427

B. C. A. (Sixth Semester) EXAMINATION, 2018

(New Course)

Paper No. BCA—N—602

ARTIFICIAL INTELLIGENCE

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meaning.

1. (a) Define artificial intelligence. Explain various characteristics of an expert system.

   (b) What is heuristic algorithm ? Enumerate various characteristics of heuristic algorithm.

2. (a) What is visual perception ? Explain its role in artificial intelligence.

   (b) What is automated reasoning ?

   (c) How can one apply intelligent behaviour of an expert system in problem solving ?

3. (a) Explain role of natural languages in expert system development.

   (b) Explain rules for Fillmore's grammar.

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4. (a) Write a program in PROLOG to add two numbers.

   (b) What are various first order predicate calculus ruler ? Enumerate and explain.

5. (a) Explain Minsky frames.

   (b) What are various rules for forward and backward deductions ? Explain.

6. (a) What do you mean by self explaining system ? Explain.

   (b) Explain the following expert system :

   (i) DENDRAL

   (ii) MYCIN

7. (a) What is pattern recognition ? Explain.

   (b) Explain the following applications of Expert system :

   (i) Speech Recognition

   (ii) Symbolic Description

8. Write short notes on any *four* of the following :

   (i) LISP

   (ii) Semantic Model

   (iii) Horn Clauses

   (iv) Grammar free analyzers

   (v) Minskey frames

   (vi) Self-explaining system

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